

Juan David Muñoz Bolaños

Rechenauerstraße 42, Innsbruck, Österreich

+43 676 9115145

jmunozbolanos@gmail.com

April 27, 2026

UltraFast Innovations

info@ultrafast-innovations.com

Application for INSTRUMENTAL SOFTWARE ENGINEER – Optical Systems

Date: April 27, 2026

Dear Hiring Manager,

I am writing to express my strong interest in the Instrumental Software Engineer position at UltraFast Innovations. With over five years of experience developing software and hardware architectures for high precision scientific instrumentation, I am eager to bring my expertise in modular software design, data acquisition, and optomechanical systems to your interdisciplinary team.

In my current role at the Medical University of Innsbruck, which I will conclude with my PhD in June 2026, I engineer complex optical metrology setups, including confocal and multiphoton microscopes. A core aspect of my work involves developing robust software and Machine Learning algorithms using Python and C++. I routinely conceptualize and implement modular software architectures that interface directly with hardware components, translating theoretical data acquisition principles into reliable, real world measurement systems. My experience encompasses object oriented design (OOD), version control via Git, and creating custom graphical user interfaces (GUIs) to gather and analyze experimental user needs effectively.

My industrial experience at INTECOL S.A.S. further solidifies my ability to deploy automated systems into demanding environments. I have successfully integrated optoelectronics, sensors, and spatial light modulators, defining system level tests and verifying performance in both laboratory and live manufacturing settings. I thrive in fast paced, project oriented environments and actively collaborate with cross functional teams to troubleshoot equipment problems and streamline R&D initiatives.

UltraFast Innovations' mission to push the boundaries of attosecond science and ultrafast photonics aligns perfectly with my ambition to build next generation scientific instruments. I am a Colombian citizen with a valid work permit for Austria, already firmly established in the region.

I welcome the opportunity to discuss how my proactive approach to system debugging and software architecture can support your technical goals. Thank you for your consideration.

Mit freundlichen Grüßen,

Juan David Muñoz Bolaños

Optical Metrologist

Juan David Muñoz Bolaños

Process & Metrology Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

jmunozbolanos@gmail.com | LinkedIn Profile



Summary

Process & Metrology Engineer with expertise in precision optical metrology, data driven process control, and machine vision. Background in developing software pipelines for automated image processing and conducting Measurement System Analysis (MSA). Ability to bridge experimental physics to precise optical measurements.

Professional/Research Experience

Jan 2022 to Present **Med. Univ. Innsbruck** *Innsbruck, Austria*
Systems Engineer | PhD Researcher

Optical Metrology & Process Monitoring: Engineered custom optical confocal microscopes and 3D tomography setups; conducted off-axis and on-axis holography, as well as interferometry, for high precision measurements. Verified system performance in laboratory environments and restored diffraction limited performance using Adaptive Optics (AO) and wavefront shaping methods.

Software Development & System Analysis: Conceptualized modular software architectures and developed Machine Learning algorithms using Python and C++. Defined system level tests, created automated measurement programs, and conducted Measurement System Analysis (MSA) to ensure reliable benchmarking.

Mar 2018 to Mar 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision System Engineer

Machine Vision & Image Processing: Engineered automated vision inspection systems for production lines using Halcon image processing and bright field/dark field illumination setups. Achieved 96% inspection reliability for label alignment, liquid level, and cap sealing, contributing to overall yield improvement.

Process Integration & Collaboration: Integrated sensor arrays and Cognex vision systems into live manufacturing environments, including automated robotic guidance. Collaborated with production and maintenance teams to evaluate process improvements and resolve equipment problems.

Technical Competencies

Software Engineering & Image Processing: Proficiency in modular software architectures, Object Oriented Design (OOD), and GUI development using Python and C++. Experience with version control (Git), machine learning classification (Python SpectraMap PyPI package), and custom phase reconstruction algorithms.

Process Engineering & Precision Metrology: Expertise in optical metrology, interferometry, holography, 3D coordinate measuring systems, and Measurement System Analysis (MSA). Advanced problem solving skills (8D, Six Sigma, root cause analysis) focused on increasing process yield, product robustness, and evaluating process simplifications to reduce manufacturing costs.

Hardware Integration & Equipment Troubleshooting: Hands on expertise in integrating optoelectronics, sensors (CCD, FLIR, SICK), and spatial light

modulators. Strong background in hardware and software interfaces (LabVIEW FPGA, C++), applying data acquisition principles (sampling theory, noise) to resolve equipment problems and transfer technologies to production.

Education

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>Doctor of Philosophy in Optical Metrology</i>	<i>Innsbruck, Austria</i>
2019 to 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	<i>Jena, Germany</i>
2012 to 2018	University of Cauca <i>Bachelor of Science in Physics Engineering</i>	<i>Popayán, Colombia</i>

Publications/Technical Evidence Awards

Awards: Selected for ZEISS SMT, ASML, and SPIE summer schools. COLFUTURO 2019 (best Colombian professionals).

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working Style: Ability to adapt to dynamic project requirements and take on new tasks; strong focus on bringing conceptual ideas to practical applications with a high degree of responsibility.

Teamwork & Process Analysis: Team oriented collaboration across optics, hardware, and software disciplines to optimize measurement workflows; structured root cause analysis of system defects and implementation of continuous improvements.

Personal Interests

- Mountain Sports (Biking, Hiking, Skiing)
- Bachata Dancing
- Harmonica
- Learning German

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

PD Dr. Christoph Krafft
Leibniz IPHT
christoph.krafft@leibniz-ipht.de

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
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Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

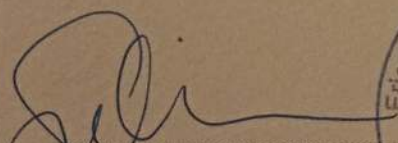
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

ASML

PhD Master Class

You will be joining us live at ASML HQ in Veldhoven on Friday and Saturday. Make sure you bring your **valid ID** to enter the ASML buildings. Without it, you will **not** be allowed inside. On Thursday we will meet at the Philips Stadium in the center of Eindhoven for an informal start to the event. The dress code for the entire masterclass is informal.

Program

Below is the program for the three days again. On the next pages, you will find more details on the different parts of the program.

Dear participant of the ASML PhD Master Class | Spring edition 2026,

We're looking forward to meeting you on March 19, 20 & 21.

Here's an information package containing details about the masterclass program. Please read this carefully so you know what to expect and how to prepare for it.

To get the most out of this masterclass for you, we want your input on what you would like to learn. At the end of this briefing, there is a link where we ask you to share your preferences with us.

Do not hesitate to contact us if you still have questions after this briefing.

Laura de Swart
Program Manager, ASML
+31 6 1562 7555

PhD Master Class

Thursday, March 19

19:30 – 22:30 Informal activity @ Philips Stadium

Friday, March 20

08:30 – 09:00	Welcome
09:00 – 10:30	Introduction into ASML Technology
10:30 – 10:50	Break
10:50 – 12:00	Pitches from former participants
12:00 – 13:00	Lunch break
13:00 – 18:00	Working on cases
18:00 – 21:45	Dinner & presentations & feedback on case results
21:45 – 22:00	Closing day one

Saturday, March 21

08:30 – 09:00	Welcome back
09:00 – 12:50	Carousel • Matching conversations • HR corner • Q&A • ASML tours
12:50 – 13:50	Lunch break
13:50 – 14:00	Wrap up and closing

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry
10 DIC/DEC 2028

Autocidad / Authority

G. ANTIOQUIA

Forma dei bulbi: / *Figures 3 and 4*

Dear Family

[illegible]

Stadtmagistrat Innsbruck
MA II - Bezirks- und Gemeindeverwaltung
Standesamt und Personenstandsangelegenheiten
Aufenthaltsangelegenheiten
post.aufenthaltsangelegenheiten@innsbruck.gv.at

Zahl: Maglbk/58566/AUFG-NAG/3/1 – 00053337-04

Antragsprotokoll

Herr: **Muñoz Bolaños Juan David**
wohnhaft in: **6020 Pradl, Reichenauer Straße 42/Top 5**
geboren am: **04.06.1994**
Reisedokument-Nr.: **AV410884**
Nationalität: **Kolumbien**
hat am: **13.03.2026**

beim Stadtmagistrat Innsbruck, Referat Aufenthaltsangelegenheiten, einen Zweckänderungsantrag auf Rot-Weiß-Rot - Karte plus eingebracht.

Ein rechtzeitig eingebrachter Verlängerungsauftrag räumt während der gesamten Dauer des Verlängerungsverfahrens dieselbe Rechtsposition ein, die nach Erhalt des letzten Aufenthaltstitels innegehabt wurde. Dies bedeutet auch, dass bei befristet erteilten Aufenthaltstiteln mit unbeschränkten Zugang zum Arbeitsmarkt („Rot-Weiß-Rot Karte plus“ und „Familienangehöriger“) weiterhin während der gesamten Dauer des Verlängerungsverfahrens ein Zugang zum Arbeitsmarkt besteht. Bei darüber hinaus gehenden Fragen einer allfälligen weiterhin bestehenden Beschäftigung wenden Sie sich bitte an das Arbeitsmarktservice (AMS).

Nachzureichende Unterlagen: ***

Die Erledigung des vollständig eingebrachten Antrages wird voraussichtlich innerhalb von Wochen erfolgen.

Persönliche Abholung (erforderlich !) jedenfalls erst ab: 17.05.2026, Zimmer: .Nummer ziehen

Zur Abholung (Abholzeiten: Montag bis Freitag, 08.00 bis 11.30 Uhr) sind mitzubringen:

- **der Reisepass**
- **Aufenthaltstitel in Kartenform**

Für den Bürgermeister:

(Huter)

ausstellende Behörde:

Stadtmagistrat Innsbruck
Abteilung II
Aufenthaltsangelegenheiten

übernommen am: 13.03.2026

Unterschrift:

Bei überprüfenden Amtshandlungen drei Monate nach Ausstellung dieser Bestätigung wird um Rückfrage bei der zuständigen Aufenthaltsbehörde ersucht. (Tel.: 0512/5360-1030/1032/1034/1036)